Name: _____ Group: _____

Fixing a Failed Separation — Resolution Triage

Team roles for today (rotate from last session): Manager keeps the group on the current question — no running ahead to the simulator. Recorder writes the group's predictions and reconciliations on this sheet. Presenter reports to the class at the checkpoints. Reflector notes where the group's thinking changed.

How this is graded: full credit requires three things for every numbered question — your **written prediction before touching any control**, the **observed result**, and, where they disagree, a sentence on **what changed in your thinking**. A correct answer with no prediction earns partial credit; a wrong prediction honestly reconciled earns full credit.

Getting started

Open drylab and load the scenario — either pick **"Ex 3 — Resolution triage"** from the *Classroom preset* menu in the HPLC sidebar, or use the link your instructor gave you (it ends in `?hplc=ex3`).

You are looking at a two-compound separation that has failed. Open the **Results** tab and record the starting point:

Quantity (from the Results tab)	Value
Resolution Rs (vs prev), second row	
Selectivity alpha (vs prev)	
Plates N (System table)	
k′ of the later peak	
Retention time of the later peak (s)	
Backpressure (bar)	

Q1. A separation needs Rs ≥ 1.5 to be quantifiable. In one sentence: what does the chromatogram *look like* that the number Rs is capturing?

The three levers

From lecture, resolution factors into three independent-looking terms:

> $R_s = (\sqrt{N}/4) \cdot ((\alpha - 1)/\alpha) \cdot (k'/(1 + k'))$

Your instructor will assign your group **one** budget line. You may only touch the controls on your line:

Team	Controls you may touch	Controls you may NOT touch
N-team	column length, particle size, flow rate	%B, temperature, solvent B
k-team	Solvent B (%), temperature	length, particle size, solvent identity
α-team	Solvent B identity (MeCN ↔ MeOH), then %B to re-tune	length, particle size

Q2 — Predict first (nothing gets moved yet). As a group, write what your lever will do to each of the four quantities when you push it hard enough to *fix* the separation. Circle a direction (↑ / ↓ / ≈) and give a one-line reason for each:

Quantity	Prediction (↑ / ↓ / ≈)	Why
Rs		
Retention time of the last peak		
Backpressure		
Which term of the equation you're moving		

STOP — show your predictions to the instructor before running.

Q3 — Now run it. Adjust only your team's controls until $R_s \geq 1.5$, or until you conclude your lever cannot reasonably get there. Record your best attempt:

Quantity	Your best attempt
Settings you changed (list each, with values)	
Rs achieved	
Retention time of the last peak (s)	
Backpressure (bar)	

Q4 — Reconcile. Where did the outcome differ from your Q2 prediction? What in your model of the separation changed?

Class checkpoint — ranking the levers

Each Presenter reports: settings, Rs, time, pressure. Fill in the class results:

Team	Rs achieved	Time cost	Pressure cost	Verdict
N				
k				
α				

Q5. One lever fixed the separation for almost nothing; one fixed it at a price no real pump would survive. Using the *form* of the resolution equation (which term is under a square root? which term saturates?), explain why the ranking came out the way it did.

Q6. The k-team likely noticed that alpha moved too, not just k′. Does that break the claim that the equation has three *independent* levers? What kind of statement is the resolution equation — a law about knobs, or a decomposition of a number?

The diminishing-returns probe (k-team's encore, everyone watches)

In the **Sweeps** tab, set the sweep to `solvent_percent` → `k_prime`.

Q7 — Predict: sketch the shape you expect for the factor k′/(1 + k′) as k′ grows from 0 to 20. At what k′ is 90 % of the maximum reached?

Q8 — Run and conclude: using the sweep and the factor you sketched, state the practical rule: past roughly what k′ does added retention buy time instead of resolution?

Homework (individual, bring the CSV)

From the **Export** tab at the *original preset condition*, download the compound-results CSV — note it carries an `operating_point` column; every number you report below must be traceable to it.

H1. Using N (System table), α, and k'_2 from your starting-point table, compute Rs from the resolution equation. Compare with the table's Rs. Report both numbers and the percent difference.

H2. Predict Rs and the last-peak retention time if the column were lengthened from 100 mm to 300 mm, using proportionality arguments only (state them). Then set the length slider to 300 mm and record what the simulator gives. Reconcile.

H3. In two sentences: when is tripling the column length the right *first* move for a failed separation — and what should be tried first?